

Instructions

- Answer all eight questions. Show sufficient working for method marks.
- A calculator may be used unless a question states otherwise.
- Write clearly in the spaces provided. Additional pages may be attached.

SCORE SAVED TO COURSE PROGRESS

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1

Explain the difference between a strong acid and a concentrated acid, with an example of each.

[5 marks]

Method sound; final simplification omitted.

5/5

2

Describe a method to prepare a pure, dry sample of a soluble salt from an insoluble base.

[6 marks]

Check the condition before applying the rule.

6/6

AGENT ANNOTATED

3

State Le Chatelier's principle and apply it to an exothermic equilibrium as temperature rises.

[5 marks]

Correct and clearly justified.

4/5

4

Calculate the number of moles in 12.0 g of carbon-12, showing the relationship you use.

[4 marks]

$$n = \frac{m}{M}$$

Method sound; final simplification omitted.

3/4

5

Balance the equation and identify the limiting reactant when 4 mol of each reactant is supplied.

[5 marks]



Check the condition before applying the rule.

5/5

AGENT ANNOTATED

6

Describe how you would determine the empirical formula of a compound from its percentage composition.

[6 marks]

Correct and clearly justified.

5/6

7

Explain, in terms of collision theory, why increasing temperature increases the rate of reaction.

[5 marks]

Method sound; final simplification omitted.

5/5

8

Determine the concentration of a solution made by dissolving 5.85 g of NaCl in 250 cm³ of water.

[5 marks]

$$C = \frac{n}{V}$$

Check the condition before applying the rule.

4/5